

Community Audio Broadcast — Engineering Solution

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Technical design for delivering live audio — five daily adhan and a weekly announcement — from one institution to 500 households, built to run unattended and to replicate across 100 institutions.

Design principles and constraints

Six constraints bind this system. Every design decision in this document traces back to one of them, and where two conflict the higher one wins.

#	Constraint	Consequence for engineering
C1	No licensed spectrum is available	All transport rides the public internet and the phone network. No RF link is designed, built or shipped
C2	₹300 per household, one-time	The box has roughly ₹285 of BOM. Every component choice is a cost argument before it is a technical one
C3	35 broadcasts a week, unattended	Scheduling is a first-class subsystem, not a cron job. Nothing may require a human at 4:45 am
C4	No technician on site, ever	Zero-touch provisioning, remote diagnosis, and swap-not-repair field policy
C5	The least-equipped household sets the floor	Every feature must degrade to something an 80-year-old with a feature phone can use
C6	100 institutions in 2 years	Multi-tenant from the first line of code; per-institution setup must cost hours, not weeks

Principles

Fail loud, never silent. The catastrophic failure in this system is a box that has quietly stopped working and a household that assumes there was no adhan today. Every endpoint reports its own health, and a device that stops reporting raises an alert. This is worth more engineering effort than audio quality.

The institution owns the audio; the platform owns nothing. The encoder captures what the microphone hears. There is no editing, no remixing, no content pipeline. This keeps the platform out of every theological and editorial argument.

Degrade, do not stop. Stream down means fall back to voice. Timetable service unreachable means run the last known schedule locally. Fixed line down means 4G. Nothing in the chain is allowed to have a single point where the household hears nothing and nobody knows.

Boring, replaceable parts. Icecast, SIP, Opus, HTTP, an ESP32. No proprietary protocol, no vendor SDK in the critical path, no component with one supplier. At 100 institutions the cost of a forced migration exceeds the cost of having built it portable.

Cheap at the edge, smart in the middle. The box does as little as possible — connect, decode, play. Timetables, retries, volumes, group membership and policy all live server-side, so a firmware bug never strands 500 devices in the field.

Explicit non-goals

- Audio fidelity beyond intelligible speech. This is a voice channel, not a music service.
- Two-way audio, presence, or any interactivity from the household.
- On-device intelligence. The box has no microphone, no sensors and no local decision-making beyond retry and reconnect.
- Offline-first operation. A household with no connectivity at all belongs on the phone tier, not on a box with a cache.

Reference architecture

Five subsystems. The line between institution-side and platform-side matters: everything institution-side must keep working for at least an hour with the platform unreachable.

```

flowchart TD
    subgraph INST[At the institution]
        MIC[PA mixer<br/>line out] --> ENC[Encoder<br/>appliance]
        CON[Console] --> ENC
        SCHED[Local schedule<br/>cache] --> ENC
    end
    ENC --> RELAY[Stream relay<br/>+ CDN]
    subgraph PLAT[Platform]
        RELAY
        API[Control API<br/>+ roster]
        TT[Timetable<br/>service]
        GW[Voice gateway]
    end
    API --> ENC
    TT --> SCHED
    RELAY --> T1[T1 app / PWA]
    RELAY --> T2[T2 Parish Box]
    RELAY --> GW
    GW --> T3[T3 phone call]

```

Subsystems

Subsystem	Runs where	Owns	Must survive
Origination	Institution	Audio capture, encoding, local schedule cache, console	Platform unreachable for 1 h; mains out for 90 min
Transport	Cloud + CDN	Stream ingest, fan-out, token gating	One relay node loss; a CDN region failing
Endpoint — box	Household	Connect, decode, play, report health	Wi-Fi loss, power loss, 30 days without an update
Endpoint — voice	Cloud	Dialling, bridging, retry, DND and time-band enforcement	Carrier rate limits; 40% no-answer rate
Control plane	Cloud	Roster, consent, timetable, policy, telemetry, billing	Region failure with no loss of enrolment data

Boundaries that matter

The encoder must be able to broadcast with the control plane down. It holds a local copy of the next 30 days of schedule and the stream ingest credentials. If the API is unreachable, the slot still opens, the stream still flows, and telemetry queues up for later. Losing the platform must not silence a masjid.

The box never talks to the institution directly. Everything goes through the platform, so a household in a different building, a different town or a different country behaves identically. This costs a few milliseconds and buys the entire multi-tenant model.

The voice gateway is a separate blast radius. It is the only subsystem with per-message cost and regulatory exposure. It runs behind its own budget cap and its own policy engine, and it can be disabled entirely without affecting the other two tiers.

Technology choices

Layer	Choice	Why, and the alternative rejected
Codec	Opus, 32 kbit/s mono, 20 ms frames	Best speech quality per bit, royalty-free, decodes on an ESP32. AAC rejected on licensing; MP3 on efficiency
Stream	Icecast-compatible HTTP over TLS	Trivially simple client, works on the box and in a browser. LL-HLS used for the app where CDN fan-out matters. WebRTC rejected: sub-second latency is not needed and per-peer cost is
Box SoC	ESP32-C3 or -S3	Wi-Fi, I2S, Opus decode in software, ₹95 at volume, enormous ecosystem
Voice	SIP trunk via an Indian DLT-registered aggregator	Regulatory registration is the hard part, not the protocol
Backend	Any boring stack; Postgres for roster and telemetry	Nothing here is novel. Do not invent

Origination subsystem

The institution already has a working audio chain. Engineering taps it rather than replacing it, which removes the single biggest source of installation risk: a muezzin or priest whose microphone now sounds different.

Audio chain

The PA mixer's unbalanced line output (or its record/tape out, where present) feeds a USB audio interface. This is a passive tap: nothing about the loudspeaker path changes, and unplugging the appliance leaves the PA exactly as it was. Where the mixer has no spare output, a 1:1 isolation transformer splits the amplifier input.

Stage	Setting	Note
Input	Line level, −10 dBV, unbalanced	Ground-loop isolation transformer fitted as standard; hum is the commonest field fault
Sample rate	48 kHz capture, resampled to 16 kHz	Speech occupies nothing above 8 kHz
Gain	Automatic gain control, 12 dB range, 200 ms attack, 2 s release	Muezzin distance from the mic varies enormously between people
Gate	−45 dBFS threshold, 30 s hang	Drives the automatic slot close
Limiter	−3 dBFS ceiling	Prevents clipping when a speaker gets close to the mic
Encode	Opus 32 kbit/s mono, 20 ms frames, VBR off	Constant bitrate keeps the box's buffer maths simple

Encoder appliance

A fanless single-board computer, sealed, wall-mounted inside the institution's existing equipment cupboard. No moving parts, no display, no keyboard. It runs three things: the capture and encode pipeline, the local schedule cache, and a watchdog that restarts the pipeline and reboots the board if the stream stalls.

- **Local autonomy.** 30 days of schedule and the ingest credentials are held on disk. A slot opens on local time whether or not the platform is reachable.
- **Network.** Ethernet to the fixed line as primary; a 4G router on a second interface as failover, with the encoder holding both routes and switching on ingest failure within 15 s. The stream drops for a few seconds and resumes; it does not end the session.
- **Watchdog.** Hardware watchdog on the board, plus a platform-side dead-man check: if the encoder has not checked in for 10 minutes, the institution's operator and the support desk are both alerted.
- **Time.** NTP-synchronised, with an RTC module so that a cold boot after a long outage still knows the time. Prayer slots depend on this absolutely.

Console

A 10-inch Android tablet in a locked wall mount running a single kiosk app, or — cheaper and preferred where the operator is uncomfortable with screens — an illuminated arcade button wired to a GPIO pin. Both do the same thing: open a broadcast, close a broadcast, show whether it is live.

The tablet adds a live connected count, a health strip that turns amber before anything has actually failed, and the week's schedule. It does not add a settings screen; everything configurable lives in the office web console, not on the wall.

Parts list

Item	Spec	₹
USB audio interface	Stereo line in, 48 kHz, class-compliant	2,200
Isolation transformer and cabling	1:1, shielded, with connectors	900
Encoder board	Quad-core SBC, 4 GB RAM, industrial SD, RTC	9,500
Enclosure, DIN mount, surge protection		3,400
4G failover router	Cat-4, external antenna	6,500
UPS	600 VA, 90 min on the encoder and router alone	6,000
Console tablet and locked mount	10-inch, kiosk-locked	12,000
Console button variant	Illuminated arcade button, GPIO	<i>1,500 alternative</i>
Total, tablet variant		₹40,500

Field faults to design for

In order of expected frequency: mains failure, ground-loop hum from the mixer tap, someone unplugging the appliance to charge a phone, the fixed line failing without the 4G SIM having credit, and the RTC battery dying. All five are handled by hardware choice or alerting, not by hoping.

Transport and latency budget

The target is **≤ 4 seconds glass-to-ear on a scheduled adhan** and **≤ 8 seconds on an announcement**, with no more than 3 seconds of spread between any two endpoints in the same neighbourhood. The spread matters more than the absolute figure: two boxes in adjacent houses a few seconds apart is what people notice and complain about.

Latency budget, adhan path

Leg	Budget	Notes
Capture and encode	60 ms	20 ms frames, three-frame encoder buffer
Institution uplink to relay	150 ms	Domestic fibre; 4G failover adds ~120 ms
Relay processing and fan-out	100 ms	No transcoding on the hot path
Relay to endpoint	150 ms	Within India, same-region edge
Client jitter buffer	1,500 ms	The single largest term, and the one that buys reliability
Decode and DAC	40 ms	
Total	≈ 2.0 s	Leaves 2 s of headroom against the 4 s target

The jitter buffer is tunable per institution and per tier. Households on poor connections get a larger buffer and accept more delay; the trade is made server-side and pushed as policy, never chosen by the household.

Keeping the spread tight

Every audio frame carries the origination timestamp. Each endpoint reports its actual playout offset in telemetry. The platform computes the spread across an institution's fleet after every broadcast, and where the spread exceeds 3 s it raises the slower endpoints' buffers to match rather than lowering the faster ones — a uniform 2.5 s is better than a ragged 1–4 s.

For the households that also hear the outdoor loudspeaker, a per-household delay offset of 0–15 s is applied at the endpoint, set once at install by ear. This is a deliberately crude mechanism; it is also the only one that actually works, because the acoustic delay depends on how far the house is from the horn.

Protocol

Icecast-compatible HTTP streaming over TLS is the primary path. It is a single GET that returns an endless Opus stream, which is about as simple as a client can be — important when that client is an ESP32 with 400 KB of usable RAM. The app and PWA use LL-HLS through a CDN instead, because browser media stacks handle it natively and CDN fan-out is where bandwidth gets cheap.

Bandwidth and cost

	Per household	500 households	100 institutions
Minutes per week	155	—	—
Data per month at 32 kbit/s	160 MB	80 GB	8 TB
Data per year	1.9 GB	≈ 1 TB	≈ 100 TB

At 100 institutions this is roughly 100 TB a year of egress. Served from origin on a VPS this is expensive; served through a CDN with an Opus stream that caches badly it is worse. The answer is a relay tier in each of two Indian regions with direct peering, and LL-HLS segments — which do cache — for the app tier, which is where most of the volume sits. Budget this at roughly **₹1.2 lakh a year at 100 institutions**, or about ₹100 per institution per month.

Access control

The stream is not public. Each endpoint holds a device token bound to a household record, exchanged at connect time for a short-lived stream URL scoped to that institution. Tokens are revoked on opt-out, on device swap, and on a household leaving the roster. A leaked URL expires within the hour.

This is deliberately light. The content is a call to prayer, not a secret; the point of gating is that the institution can say truthfully that it broadcasts to its own members and not to the public, which is what keeps the whole system outside broadcasting regulation.

Parish Box — hardware

One knob, one light, one job. The box is a radio that only receives one station, and every design decision protects that simplicity and the ₹285 target.

```
flowchart LR
```

```
PSU[5V 1A<br/>adaptor] --> REG[3.3V LDO]
```

```
REG --> SOC[ESP32-C3<br/>WiFi + I2S]
```

```
SOC --> DAC[I2S DAC<br/>or PWM]
```

```
DAC --> AMP[3W class-D]
```

```
POT[Volume pot] --> AMP
```

```
AMP --> SPK[66mm<br/>3W speaker]
```

```
SOC --> LED[Status LED]
```

```
BTN[Reset/pair<br/>button] --> SOC
```

Component selection

Block	Choice	Why this, not the obvious alternative
SoC	ESP32-C3-MINI-1	Wi-Fi 4 and enough headroom to decode Opus at 32 kbit/s in software. An ESP32-S3 is used if DSP headroom proves tight in prototyping; a Pi Zero W is four times the price
Audio out	Internal DAC via I2S to a discrete PCM5102-class chip on the Pro build; PWM through an RC filter on the cost build	The PWM path saves ₹24 and is acceptable for speech. Prototype both and decide by listening, not by spec sheet
Amplifier	PAM8403-class class-D, single channel	Efficient enough to run off the same 5 V rail without a heatsink
Speaker	66 mm, 3 W, 4 Ω, paper cone, full range	Voice-band response matters; bass does not. A larger driver adds cost and enclosure volume for no intelligibility gain
Power	External 5 V 1 A BIS-marked adaptor	Keeps mains entirely outside the enclosure. No mains creepage, no safety certification burden on our PCB, and a failed adaptor is a ₹52 swap
Storage	On-SoC flash, 4 MB	Holds firmware, the A/B slot, and a single chime. No SD card — SD cards are the second commonest field failure in cheap consumer audio

Bill of materials at volume

Item	500 units	5,000 units	50,000 units
ESP32-C3-MINI module	95	82	71
Class-D amplifier	14	11	9
Speaker, 66 mm 3 W	32	26	21
Enclosure, ABS, tooled at 5k+	42	30	19
5 V 1 A adaptor, BIS	52	44	38
PCB, pot, LED, button, passives, wiring	28	21	16
Assembly, flashing, functional test	22	16	11
BOM total	₹285	₹230	₹185
Add: tooling amortised	—	₹12	₹1
Landed cost	₹285	₹242	₹186

At 500 units the enclosure is off-the-shelf and the margin to ₹300 is ₹15. A tooled enclosure needs roughly **₹1.8 lakh of mould cost** and only pays back past about 4,000 units — which is why the first hardware decision at scale is when to commit to tooling, not what to put inside the box.

Enclosure and acoustics

Wall-mountable and shelf-standable, ported front grille, drip-resistant but not sealed. Roughly 110 × 80 × 45 mm. The volume knob is a physical rotary with a detented off position, because an off position that is visibly off is what stops a frustrated household from unplugging the box and losing it from the fleet permanently.

One bicolour LED: steady green ready, slow green pulse playing, amber cannot reach Wi-Fi, red no power path fault. Colour is paired with pulse pattern so it reads for colour-blind users.

Power

State	Draw	Duration per day
Standby, Wi-Fi modem sleep	0.35 W	~23.7 h
Playing	1.8 W average, 3 W peak	~15 min at five slots

State	Draw	Duration per day
Daily energy	≈ 8.8 Wh	≈ 3.2 kWh/year, about ₹30

No battery. A battery adds ₹60 of BOM, a charge controller, a disposal problem and a component that fails inside three years. A household whose power is out is a household that is not listening anyway, and the box resumes automatically when power returns.

Certification

The box is a low-power 2.4 GHz device using a pre-certified radio module, which is the whole reason for choosing a module over a bare chip. It needs WPC ETA self-declaration based on the module's existing approval, BIS registration for the adaptor (bought pre-certified), and RoHS declarations from suppliers. Budget **₹85,000 and eight weeks** for the first certification pass; it does not repeat per batch.

Parish Box — firmware

The firmware's job is to be boring for five years. It holds no schedule, no policy and no content; it connects, plays what it is told, and reports what happened.

```
stateDiagram-v2
    [*] --> Boot
    Boot --> Provision: no credentials
    Boot --> Connecting: credentials present
    Provision --> Connecting: volunteer sets Wi-Fi
    Connecting --> Standby: registered
    Connecting --> Offline: retries exhausted
    Offline --> Connecting: backoff expires
    Standby --> Playing: play command
    Playing --> Standby: stream ends
    Playing --> Reconnect: stream lost
    Reconnect --> Playing: recovered
    Reconnect --> Standby: gave up
```

Standby and wake

In standby the box holds a long-lived authenticated connection to the platform in Wi-Fi modem-sleep, waking on a keepalive every 30 seconds. A play command names the stream URL, the volume for this slot, and the playback delay offset. There is no local clock decision and no polling of a schedule: if the platform says play, it plays.

This is a deliberate trade. It means a box cannot play a slot while the platform is unreachable — unlike the encoder, which can. It is the right trade because a box that decides for itself when to make noise at 4:45 am, using a clock that may have drifted, is a much worse failure than silence.

Provisioning

A volunteer installs the box in under three minutes, with no app on their own phone and no account:

1. Power on. The box finds no credentials and raises a Wi-Fi access point named for the institution and the device's short code.
2. The volunteer joins that network from any phone; a captive portal opens automatically.
3. The portal lists nearby networks. The volunteer picks the household's, types the password, taps save.
4. The box joins, registers itself against the pre-assigned household record using its factory device identity, and plays a short confirmation chime.
5. The volunteer taps "test" in the office web console; the box plays a five-second test tone. Install verified, on the record, before the volunteer leaves.

The device identity and the household binding are created at the office **before** the visit, so the volunteer never types a household name or a serial number in the field. Re-provisioning after a Wi-Fi password change repeats steps 1–4 by holding the pair button for five seconds.

Over-the-air update

A/B partitions with rollback. An update is fetched in standby only, verified by signature, written to the inactive slot, and booted once. If the new firmware does not register with the platform within 90 seconds, the bootloader falls back to the previous slot permanently and reports the failure.

Updates are staged: 1% of the fleet, then 10%, then the rest, with a 48-hour soak at each step and an automatic halt if the failure rate of any stage exceeds 2%. Updates never run inside a scheduled slot window, and never between 22:00 and 06:00 local.

Failure behaviour

Condition	Behaviour
Wi-Fi lost	Amber LED, exponential backoff to a 5-minute ceiling, reconnect forever

Condition	Behaviour
Stream lost mid-slot	Reconnect up to 3 times within 10 s, then return to standby silently — never a burst of noise
Platform unreachable at slot time	Box stays silent; platform-side telemetry gap raises the alert
Power cycled	Resumes to standby automatically, no household action
Corrupt firmware	Rollback to previous slot, report, keep working
Household turns volume to zero	Detected as a distinct state and reported — this is an early signal of R16, a household about to disengage

What the firmware reports

Per slot: joined or not, join latency, actual playout offset, buffer underruns, RSSI, volume position, uptime, firmware version. Nothing about the household, no audio, no microphone — the box has none. This is the minimum needed to detect a silent failure and the maximum that can be justified under data minimisation.

Scheduler and timetable

This is the subsystem that turns a broadcast tool into something that runs 35 times a week without anyone touching it. It is also where a bug is most publicly embarrassing: adhan at the wrong time, every day, in front of the whole community.

Computing prayer times

Times derive from the institution's latitude, longitude and elevation, the date, and two configuration choices the institution makes once:

Setting	Options	Note
Calculation method	Muslim World League, Umm al-Qura, Karachi, Egyptian, and others	Sets the solar depression angles for Fajr and Isha
Asr madhhab	Standard (Shafi'i, Maliki, Hanbali) or Hanafi	Changes Asr by roughly 45–90 minutes

Setting	Options	Note
High-latitude rule	Not applicable in India	Kept in the model for portability
Per-slot offset	−60 to +60 minutes, per prayer, permanent	This is how a masjid encodes its actual local practice

The computation itself is well-established astronomy and should use a maintained library rather than being written from scratch. **The institution's printed timetable is the source of truth, not the computation.** During commissioning the computed times are diffed against the masjid's own timetable for a full month and the offsets adjusted until they agree. If they cannot be made to agree, the institution's timetable is imported wholesale instead.

Slot types

Slot	Typical duration	Opens	Notes
Five daily adhan	3 min	Computed + offset	The core load
Iqamah	1 min	Fixed minutes after adhan	Optional, per institution
Jumu'ah khutbah	30–45 min	Fixed weekly time	Longest regular slot
Weekly announcement	5–20 min	Operator-initiated or scheduled	
Taraweeh, sehri, iftar	Varies	Ramadan calendar	Seasonal slot set, enabled for the month
Urgent notice	Any	Operator, immediate	Overrides quiet windows

The model must handle at least 8 slots a day and a seasonal slot set that activates and deactivates on a date range. Church equivalents — Angelus three times a day, daily mass, novena — use exactly the same machinery with a different generator.

Opening and closing a slot

```
sequenceDiagram
    participant S as Scheduler
    participant E as Encoder
    participant P as Platform
    participant B as Boxes
    S->>E: arm slot T-60s
    S->>P: pre-warm fleet T-30s
    P->>B: wake, expect stream
    E->>P: stream opens at T
    P->>B: play(url, volume, offset)
    Note over E: gate detects speech
    Note over E: 30s silence OR 6min cap
    E->>P: stream closes
    P->>B: stop
```

The pre-warm at T-30 s is what makes the box appear instant: it has already connected and buffered before the muezzin speaks. Without it the first four seconds of every adhan are lost to connection setup, which is exactly the part people notice.

Resilience

- **Local cache.** The encoder holds 30 days of resolved slot times. A slot opens on local time with the platform down.
- **Nightly recompute.** Times recomputed at 02:00 local, pushed to the encoder, and the coming week shown on the console so the operator can see an error before the community hears it.
- **Clock.** RTC plus NTP. A device whose clock has drifted more than 30 seconds refuses to open a slot and raises an alert instead.
- **Missed slot alert.** If a slot opens and no audio crosses the gate within the grace period, the slot closes, and the operator and support desk are notified. A missed Fajr must be visible by breakfast.
- **Fallback recording.** Disabled by default. Where an institution enables it, a recording of its own muezzin plays after the grace period. This is a configuration flag with an audit entry, not a default behaviour, because whether recorded adhan is acceptable is the institution's call and nobody else's.

Per-household filtering

The scheduler emits one slot event; the platform fans it out per household against that household's own policy — which slots it takes, at what volume, and whether a quiet window suppresses it. A household that has switched Fajr off simply receives no play

command. All of this is server-side, which means a household's preferences can be changed by the office in seconds without touching the device.

Voice gateway

The phone tier reaches households that have nothing — no smartphone, no broadband, often no one in the house under seventy. It is also the only subsystem that costs money per minute and carries regulatory exposure, so it is engineered as a separate, capped, disableable service.

What it does and does not carry

Carries	Does not carry
The weekly announcement	The five daily adhan
Urgent notices inside 09:00–21:00	Anything outside 09:00–21:00
Replay on dial-in (Pro)	Anything to a household without recorded consent

The exclusion of adhan is not a technical limit. It is a time-band and cost decision taken upstream: automated outbound calls before 9 am rule out Fajr whatever consent exists, and five calls a day to 500 households would run to roughly ₹6.5 lakh a year against ₹19,000 for a weekly-only tier.

Architecture

A SIP trunk from a DLT-registered Indian aggregator, a media server that bridges answered calls into the live Opus stream after transcoding to G.711, and a dialler that manages the ramp. The transcode is the only place in the system where audio is re-encoded, and it costs roughly 40 ms.

```

flowchart LR
    SCH[Slot event] --> POL[Policy engine<br/>consent, time, budget]
    POL --> DIAL[Dialler<br/>ramped]
    DIAL --> TRUNK[SIP trunk]
    TRUNK --> HH[Household phone]
    RELAY[Live stream] --> MS[Media server<br/>Opus to G.711]
    MS --> TRUNK
  
```

Dial ramp and concurrency

Calling 75 households at once is not possible and not desirable. Carriers apply calls-per-second limits, and a burst looks like a robocall campaign.

Parameter	Value	Reason
Calls per second	5, negotiated with the aggregator	Above this, carriers throttle or block
Concurrent channels	40	Enough for 75 households with a 50% answer rate
Ramp to 95% of list	≤ 4 minutes	Matches the announcement's typical length
Retries	2, at +3 min and +8 min	Beyond two, answer rate does not improve and cost does
Ring timeout	25 s	Longer just fills voicemail
Late join	Allowed	A household answering at minute 6 hears the rest live, not from the start

Policy engine

Every call passes four gates before dialling, and a failure at any gate is logged with a reason:

1. **Consent.** A dated, stored consent record for this household exists and has not been withdrawn.
2. **Time band.** Local time is inside 09:00–21:00.
3. **Slot eligibility.** This slot type is enabled for the voice tier at all.
4. **Budget.** The institution's monthly voice spend cap has not been reached.

The budget gate is a hard stop, not a warning. It is the mechanism that prevents a misconfiguration — a stuck retry loop, an institution enabling adhan on the voice tier by accident — from producing a bill nobody authorised. The operator sees the month's spend against cap on the console before every broadcast.

Regulatory plumbing

DLT registration, a registered CLI header, and pre-registered templates are prerequisites, not features. The header is what households see on the incoming call, and it should read as the institution, not the platform — a call from an unrecognised number at 11 am does

not get answered. Registration takes several weeks and sits on the critical path, so it starts in the first phase of any deployment.

Open legal question. Whether a non-commercial religious notice to consenting members falls inside TCCCPR's definition of commercial communication is genuinely unsettled. The system is engineered as though it does — full consent, full time-band compliance, full DLT registration — because that is the only posture that survives if the answer turns out to be yes. This should still be put to a telecom lawyer before the first institution goes live.

Cost control

At ₹0.12 per 30-second pulse before GST, a 20-minute announcement to 75 answered households is about ₹360. The engineering levers are answer-rate measurement (do not pay to retry numbers that never answer), duration caps, and per-institution budget enforcement. The product lever is stronger: every household moved from the voice tier to a box removes its recurring cost permanently and pays back the ₹285 in under a year.

Backend platform

Multi-tenant from the first commit. Retrofitting tenancy at institution twelve costs more than building it at institution one, and the target is a hundred.

Services

Service	Responsibility	Scaling shape
Roster	Institutions, households, endpoints, consent, preferences	Small, relational, the system of record
Timetable	Compute, override, publish 30-day windows	CPU-trivial, cacheable
Broadcast orchestrator	Arm, open, fan out, close, reconcile	Bursty — 35 slots per institution per week, clustered at prayer times
Device gateway	Persistent connections to every box, play commands, telemetry ingest	The one service that scales with household count, not institution count
Voice gateway	Dial, bridge, retry, enforce policy	Separate blast radius, separate budget
Telemetry and reporting	Delivery reports, health, alerts, impact data	Write-heavy, append-only

Service	Responsibility	Scaling shape
Admin console	Office web UI, operator console API	Low traffic

The device gateway is the only piece with a genuinely interesting load profile: at 100 institutions it holds roughly **30,000 persistent connections**, almost all idle, all waking within the same few seconds five times a day because prayer times across a district are minutes apart. It must be sized for the burst, not the average, and the fan-out must be staggered over the 30-second pre-warm window rather than issued in one shot.

Data model

```

flowchart TD
    ORG[Institution] --> HH[Household]
    ORG --> SLOT[Slot definition]
    ORG --> OPR[Operator]
    HH --> EP[Endpoint<br/>box / app / phone]
    HH --> CON[Consent record]
    HH --> PREF[Slot preferences]
    SLOT --> BC[Broadcast instance]
    BC --> DEL[Delivery record<br/>per endpoint]

```

The delivery record is the highest-volume object: one row per endpoint per broadcast, so $500 \times 35 \times 52 \approx$ **910,000 rows per institution per year**, or 91 million at 100 institutions. It is append-only, partitioned by month, and rolled up nightly into per-household and per-institution aggregates. Raw rows are retained 90 days; aggregates indefinitely, because the aggregates are what grant and CSR reporting needs.

Tenancy and isolation

One database, row-level tenancy keyed on institution, enforced in a single data-access layer that no query path bypasses. Not a database per tenant — at 100 institutions that is 100 migrations to run and 100 backups to verify, for isolation that row-level security already gives.

One institution can never see another's roster, and this must be true by construction rather than by careful coding. A federation or diocese that operates several institutions gets a group role that spans them explicitly, which is a first-class concept rather than a workaround, because that is how the sales motion works.

APIs

- **Device API.** Persistent authenticated connection. Register, receive play and stop, push telemetry. Deliberately tiny — five message types.
- **Encoder API.** Fetch schedule window, fetch ingest credentials, report stream state, queue telemetry when offline.
- **Console API.** Everything the office and the operator do. Roster CRUD, bulk import, preferences, delivery reports, budget.
- **Federation API.** Read-only, aggregate-only. Lets a diocese or waqf board see adoption across its institutions without seeing individual households. This is the API that makes the partnership channel possible, and it is worth building before it is asked for.
- **Impact export.** Signed, dated aggregate reports for grant and CSR reporting. Covered in the cost-to-serve section.

What deliberately does not exist

No content management. No recording in Basic. No household-facing account system — households are identified by a device token or a phone number, never a username and password, because an elderly household that cannot reset a password is a household lost from the fleet.

Security, privacy and compliance

The asset worth protecting is not the audio — it is broadcast to a thousand ears by design. It is the ability to make 30,000 speakers in people's homes play something, and the roster of who belongs to which religious community.

Threat model

Threat	Severity	Control
An attacker injects audio into a live stream	Critical	Ingest is authenticated per institution with a rotating credential held only on the encoder; the relay accepts one publisher per mount and rejects a second
An attacker triggers playback on the fleet	Critical	Play commands originate only from the orchestrator, signed, and the device gateway is not reachable from the public internet except by registered devices

Threat	Severity	Control
The roster leaks — names, addresses and phone numbers by religious affiliation	Critical	Row-level tenancy, encryption at rest, access limited to two named office roles, every read audited
A device token is extracted from a box and used to eavesdrop	Low	Content is public-by-nature; tokens are per-device, revocable, and short-lived at the stream layer
A malicious firmware update bricks the fleet	High	Signed images, A/B rollback, staged rollout with automatic halt
Denial of service against the relay at prayer time	Medium	Two regions, CDN in front of the app tier, and the encoder's local autonomy means the institution can still fall back to its loudspeaker
An insider at an institution broadcasts something harmful	Medium	Single named originator per institution, full audit log of who opened which slot, and the same content rules that apply to the loudspeaker

The first three are the ones that matter. A community audio system that can be hijacked to play arbitrary audio into hundreds of homes in a religious neighbourhood is not a technical problem, it is a public-order problem, and it should be treated with that seriousness throughout.

Device identity

Each box carries a factory-provisioned identity written at flashing time — a device ID and a private key in the SoC's protected flash, with secure boot and flash encryption enabled. The identity binds to a household record at the office before installation. A box that is re-provisioned to a different household requires an office action, which is also the anti-theft control: a stolen box is unbound and worthless.

Data protection under the DPDP Act

The roster is personal data, and its subject matter — religious community membership — makes it sensitive in practice even where the statute does not name a special category.

Obligation	Implementation
Lawful basis	Explicit, dated, purpose-limited consent captured at enrolment, stored with its exact wording and version
Purpose limitation	The roster is used to deliver broadcasts and nothing else. No analytics product, no resale, no cross-institution profiling, stated in the contract
Data minimisation	Household name, address cluster, phone numbers, tier, preferences. No listening history beyond connected-or-not, no location, no contacts
Withdrawal	One request to the office. Endpoint tokens revoked, phone numbers suppressed, effective before the next slot
Erasure	Full deletion within 30 days on request, including from backups on their normal rotation
Retention	Raw delivery rows 90 days; aggregates indefinitely, with no household identifier
Breach	Notification path defined before launch, not after

The institution is the data fiduciary; the platform is the processor. This distinction is worth getting right in writing at the first institution, because at a hundred institutions it determines who answers when something goes wrong.

What the system must never do

No recording of household audio — the box has no microphone, and this is a hardware fact rather than a software promise, which is a much stronger statement to make to a community. No listening analytics sold or shared. No use of the roster for anything other than broadcast delivery. These should be published commitments, not internal policy, because the trust that makes this product adoptable is the only real moat it has.

Observability and field operations

A box that quietly stopped working three weeks ago, in a house where nobody thought to mention it, is the defining failure of this system. Everything here exists to make that impossible.

The three alerts that matter

Alert	Trigger	Goes to	Response
Slot missed	A scheduled slot opened and no audio crossed the gate	Operator and support desk, within 2 minutes	Operator checks the mic; support checks the encoder
Endpoint dark	An endpoint has missed 3 consecutive slots it was eligible for	Office weekly list	Field volunteer visit
Institution dark	Encoder has not checked in for 10 minutes	Support desk, paging	Remote diagnosis, then site visit

Everything else is a dashboard. These three are the only things anyone gets woken for, and the first is the only one the institution itself sees.

Per-broadcast reconciliation

After every slot the orchestrator writes a reconciliation: eligible endpoints, play commands sent, endpoints that acknowledged, endpoints that actually played, calls attempted, calls answered, and the playout spread across the fleet. This is simultaneously the delivery report the institution sees, the health signal engineering watches, and the impact data that grant reporting needs — one artefact, three consumers, which is why it is worth writing properly.

Metrics worth watching

Metric	Healthy	Meaning when it drifts
Slot open success rate	$\geq 99\%$	Scheduler or encoder problem
Endpoint join rate within 20 s	$\geq 95\%$	Pre-warm not working, or fleet-wide Wi-Fi conditions
Playout spread, p95	≤ 3 s	Buffer policy needs retuning
Buffer underruns per playing hour	≤ 2	Bandwidth or relay problem
Voice answer rate	$\geq 60\%$	Wrong numbers, or the CLI header is not recognised
Endpoints dark, rolling 30 days	$\leq 3\%$	The adoption problem showing up as a technical metric

Metric	Healthy	Meaning when it drifts
Volume-at-zero count	$\leq 2\%$	Households disengaging — the earliest warning available

The last two are product metrics wearing engineering clothes. They belong on the same dashboard because by the time they show up in a subscription renewal conversation it is far too late.

Field support

Swap, never repair. A failed box is replaced from a spares pool held at the institution, at 5% of fleet. The failed unit is returned in batches, triaged centrally, and either refurbished or scrapped. A volunteer must never be asked to open an enclosure.

Triage is a one-page sheet. Amber light means Wi-Fi — re-provision. No light means power — swap the adaptor first, it is the commonest failure and it costs ₹52. Green but silent means the household has the volume down or the slot switched off — check the console before visiting. That covers the large majority of real faults, and none of it requires a technician.

Remote first. Before any visit, support can see the endpoint's last check-in, RSSI trend, firmware version, volume position and slot history. Most "my box is broken" calls resolve without a visit, and at 100 institutions that difference is the whole field-operations budget.

Logistics at scale

	1 institution	100 institutions
Boxes in the field	400	40,000
Expected annual failures at 5%	20	2,000
Spares held	20 at the institution	20 per institution, plus 500 central
Field visits per week	~2	~40, handled by local installer partners, not central staff

At 100 institutions field support cannot be a central team, which is the engineering argument for the installer-partner channel: the same people who installed the boxes swap them, guided by remote diagnosis and the one-page sheet. The platform's job is to make the diagnosis good enough that a local electrician can act on it.

Manufacturing and supply chain

At 100 institutions this is a 40,000-unit hardware programme with a working-capital requirement of roughly **₹75 lakh**, which makes it the largest single financial commitment in the whole plan and the one most likely to be underestimated.

Volume by stage

Stage	Institutions	Boxes	Order pattern	Unit cost
Pilot	1	400	One batch, off-the-shelf enclosure	₹285
Early	10	4,000	Quarterly batches, tooling decision due	₹242
Scale	100	40,000	Monthly, forecast-driven	₹186

What changes between 500 and 50,000

Enclosure. Off-the-shelf at 500. An injection-moulded enclosure costs about **₹1.8 lakh in tooling** and pays back past roughly 4,000 units, so the decision falls at around institution ten. Tooling also fixes the design — after the mould is cut, changing the knob position costs a new mould, so the mechanical design must be settled before the order.

Assembly. Hand assembly by a local contract shop at 500. Above about 5,000 units a year this needs an SMT line with pick-and-place, which means a real EMS partner, a proper design-for-manufacture pass, and panelised PCBs.

Test. At 500, a bench jig and a person with headphones. At scale, an automated end-of-line jig: power up, verify Wi-Fi association, play a tone, measure output on a mic, write the device identity and certificate, print and apply a label. **Build the jig at the pilot stage even though it is not needed yet** — it is also how field-returned units get triaged, and retrofitting identity provisioning into an existing line is painful.

Identity. Keys and certificates written during end-of-line test, from a provisioning service, with a record of every identity issued. This is what makes the anti-theft and multi-tenant binding work, and it cannot be bolted on afterwards.

Sourcing risk

Component	Risk	Mitigation
ESP32 module	Single vendor family; historical allocation shortages	Qualify two module vendors on the same footprint; hold 3 months of buffer stock at scale

Component	Risk	Mitigation
Speaker	Commodity, many suppliers	Qualify two; acoustic performance varies more than datasheets suggest — listen before switching
Adaptor	BIS registration is per-model	Buy pre-certified; a substitution needs a new BIS check
Enclosure	Single-tool dependency after tooling	Keep the mould as an owned asset with a named alternative moulder

Design for substitution. Every component that can have a second source should have one qualified before it is needed, and the PCB should accommodate both footprints where that costs nothing. A four-week module shortage that stops 4,000 installs is a much bigger problem than a ₹3 higher unit cost.

Working capital

This is the part that catches social enterprises. Boxes are paid for on order and recovered from households or institutions months later, across a long enrolment cycle.

	10 institutions	100 institutions
Boxes ordered per year	4,000	40,000
Cash out at landed cost	₹9.7 lakh	₹74.4 lakh
Typical lag, order to recovery	4–6 months	4–6 months
Peak working capital tied up	≈ ₹5 lakh	≈ ₹37 lakh

Three levers: take deposits from institutions before ordering, fund box batches from CSR or grant capital rather than revenue, and order in monthly rather than quarterly batches once volumes justify it. The first is the most important and the easiest to forget — **an institution that has committed money before the order is placed also enrolls households faster**, so the deposit does double duty.

Repair and end-of-life

Returned units are triaged centrally. Most failures are adaptors and speakers, both of which are replaceable in minutes at a bench. A box that fails at the board is scrapped and its enclosure and speaker recovered. E-waste handling is a registered-recycler contract, needed under the E-Waste Management Rules once volumes are meaningful — budget for it from institution ten rather than discovering it at institution fifty.

Cost-to-serve and what engineering owes monetization

The subscription price is set by what an institution will pay. Whether that price works is set by cost-to-serve, and cost-to-serve is an engineering output.

Platform cost per institution per month

Line	1 institution	10	100	Scaling behaviour
Streaming relay and egress	₹1,250	₹400	₹100	Falls hard — fixed relay capacity amortises
Device gateway compute	₹600	₹150	₹45	Falls, then flattens at connection density
Database and telemetry storage	₹400	₹120	₹60	Falls, then grows with delivery-row volume
Voice minutes (weekly tier only)	₹1,580	₹1,580	₹1,580	Flat — pure variable cost, never amortises
Notifications	₹500	₹500	₹500	Flat
Support and field ops, allocated	₹5,000	₹2,000	₹900	Falls with channel partners doing field work
Monitoring, backups, security	₹800	₹200	₹70	Falls
Total per institution/month	₹10,130	₹4,950	₹3,255	

This excludes hardware, which is a separate one-time recovery, and excludes engineering salaries, which are a fixed cost the scale has to cover.

The important line is voice. It is the only cost that does not fall with scale, and at 100 institutions it is nearly half of what remains. Every household moved from the phone tier onto a box removes about ₹21 a month of platform cost permanently. Engineering's contribution to margin is not making servers cheaper — it is making the box cheap and reliable enough that the phone tier shrinks.

What this implies for pricing

At 100 institutions, cost-to-serve is roughly **₹3,250 per institution per month** all-in. An institution subscription needs to sit meaningfully above that to cover engineering, sales and the working capital tied up in hardware. Below about 30 institutions the platform

loses money on every one, which is a funding question rather than a pricing one — covered in the go-to-market document.

What engineering must build for the money side

These are not features anyone asks for in a demo, and all three are load-bearing for the business model.

Metering and billing. Per-institution usage: broadcast minutes, endpoints active, voice minutes and spend, storage. Must be accurate enough to invoice from and to enforce a hard voice budget cap. Build it before institution ten, because reconstructing usage history afterwards is not possible.

Impact reporting for CSR and grants. Funders buy verified reach, not software. The system should produce a signed, dated report per institution and per funder covering households enrolled, households reached, broadcasts delivered, reach by tier, and a coverage trend — with no household-level identifiers in it. This is a genuine competitive advantage: almost no comparable community-media intervention can evidence delivery at household granularity, and this one produces that evidence as a by-product of running normally.

Household contribution collection. Where households contribute a small monthly amount, the collection is the institution's, not the platform's — but the platform must track who has contributed so the institution's clerk is not running a parallel register. UPI collection links and reconciliation against the roster, no payment processing of our own.

Federation-level rollups. A diocese or waqf board funding twenty institutions needs one view across them, without household detail. This is the same aggregate pipeline the impact report uses, exposed through the federation API.

Cost levers engineering controls

Lever	Saving	Cost of pulling it
Shrink the voice tier by one household	₹250/year, forever	₹285 of box, recovered in 14 months
Drop bitrate 48 → 32 kbit/s	~33% of egress	Marginal quality loss, already taken
Cache LL-HLS segments at CDN edge for the app tier	~50% of app-tier egress	Adds 2–3 s latency on that tier only
Remote-resolve a field fault instead of a visit	₹400–800 per incident	Better telemetry, already being built

Lever	Saving	Cost of pulling it
Tooled enclosure at 4,000 units	₹12–55 per box	₹1.8 lakh and a frozen mechanical design

Scaling to a hundred institutions

The technical scaling is unremarkable — 30,000 idle connections and 100 TB of egress a year is a modest system. What actually breaks is everything around the software: onboarding time, field support, hardware logistics, and the assumption that one team knows every deployment.

What breaks, in order

Institutions	What breaks first	What must exist before it does
1 → 5	Every deployment is bespoke; the team attends every go-live	A commissioning runbook, and a second institution onboarded by someone who did not build the first
5 → 20	Field visits consume the engineering team	Installer-partner channel, remote diagnosis good enough for a local electrician, one-page triage sheet
20 → 50	Hardware working capital and batch logistics	Deposits before ordering, a forecast, an EMS partner, tooled enclosure, end-of-line test jig
50 → 100	Prayer-time burst on the device gateway; support ticket volume	Staggered fan-out, regional relays, self-service console that the office clerk can actually use
100 → 500	Central support model; single-region infrastructure; one-size timetable assumptions	Tiered partner support, multi-region, and a timetable model that has survived several denominations

Onboarding a new institution

This is the metric that decides whether 100 institutions in two years is real. It has to fall from weeks to about two days of platform-side effort.

Step	Pilot	Target at scale	How
Tenant creation and configuration	2 days	30 minutes	Self-service wizard: coordinates, calculation method, slot set, branding
Timetable validation against printed	1 month	1 week	Automated diff against an uploaded photo or PDF of their timetable
Encoder commissioning	1 day on site	2 hours, by a partner	Pre-configured appliance shipped ready; partner plugs in, scans a code, done
Roster import	1 week	1 day	Bulk import with validation, from whatever the institution already keeps
Box provisioning	—	Pre-bound at the warehouse	Device identities bound to household records before shipping
Total platform effort	~3 weeks	~2 days	

The single highest-leverage investment is the **pre-configured, pre-bound shipment**: an institution receives a box of boxes already tied to its household records, and a volunteer with no technical skill does the rest. Everything else on this list is ordinary product work.

Infrastructure shape at 100

- Two Indian regions, active-active for the relay tier, active-passive for the control plane.
- CDN in front of the app tier only — the box tier's Opus stream does not cache usefully.
- Device gateway horizontally partitioned by institution, so a bad deployment affects a subset rather than the fleet.
- Voice gateway isolated, with per-institution budget caps and a global circuit breaker.
- Burst sizing: assume every institution in a district has prayer times within 15 minutes of each other, so the real peak is roughly 30% of the whole fleet waking inside one minute.

None of this is exotic. It should be built to this shape early not because the load demands it, but because the alternative is re-architecting during the growth phase, which is when there is least time for it.

Variation across institutions

The hidden scaling problem is that institutions are not identical. Different calculation methods and madhhabs, different slot sets, churches with entirely different rhythms, Ramadan and Lent schedules, regional languages, and institutions whose printed timetable simply does not match any standard computation. **Every one of these must be configuration, never a code branch.** The moment there is a per-institution code path, onboarding stops scaling and the two-day target is gone.

Roadmap, team and open risks

Build order

Quarter	Build	Proves
Q1	Encoder, relay, PWA, console button, scheduler with timetable	An adhan reaches a phone on time, five times a day, unattended
Q2	Box hardware revision A, firmware, provisioning portal, OTA	A volunteer installs a box in 3 minutes and it works for a month untouched
Q3	Roster, consent, voice gateway, DLT, delivery reports	The full 500-household pilot runs 26 weeks
Q4	Multi-tenancy hardening, onboarding wizard, partner tooling, metering	A second institution onboards without the build team
Y2 H1	Tooled enclosure, EMS transfer, test jig, regional relays	10 institutions, hardware at ₹242
Y2 H2	Federation API, impact reporting, Pro tier (replay, archive, app)	Funder-grade evidence, and a reason to upgrade

Pro — replay, archive, transcripts, native app — is deliberately last. It is the upsell, and none of it matters if the Basic channel is not dependable.

Team

Role	When	Note
Embedded engineer	Q1	Firmware, hardware bring-up. The hardest hire and the one to make first
Backend engineer × 2	Q1	Platform, scheduler, device gateway

Role	When	Note
Full-stack engineer	Q2	Console, office web, provisioning portal
Hardware / manufacturing lead	Q2, part-time then full	BOM, EMS, certification, logistics
Field operations lead	Q3	Installer partners, triage, spares
SRE, part-time	Q4	Only once there is something to page about

Six to seven people through year one. The embedded engineer is the constraint — the box is the product, and a firmware bug that strands 40,000 devices is unrecoverable in a way that a backend bug never is.

Build versus buy

Component	Decision	Reason
Prayer-time computation	Buy / use library	Solved, well-tested, and getting it subtly wrong is highly visible
Streaming relay	Buy (managed Icecast or equivalent) initially, build past 20 institutions	Egress cost eventually justifies owning it
Voice / SIP	Buy — an Indian DLT-registered aggregator	The registration and carrier relationships are the product, not the protocol
Box hardware	Build	Nothing off-the-shelf hits ₹285 with this behaviour. This is the defensible part
Box firmware	Build	Same
Device management / OTA	Buy if a credible ESP32-class service exists at this price, else build thin	Do not build a full IoT platform
Billing and payments	Buy	Never build payments

Open technical risks

These need prototypes, not documents, and the first three should be settled before any hardware is ordered.

#	Risk	How to settle it
T1	Opus decode plus Wi-Fi on an ESP32-C3 may not leave enough headroom for glitch-free playback	Build it first, in week one. If it fails, move to ESP32-S3 at about ₹25 more and re-check the ₹300 ceiling
T2	PWM audio output may not be acceptable for adhan, which is melodic rather than pure speech	Blind listening test with actual community members, not engineers. ₹24 rides on the answer
T3	Household Wi-Fi in the target segment may be far worse than assumed — weak signal, frequent password changes, cheap routers dropping idle clients	Instrument 25 boxes for a month in Phase 1 before committing to volume
T4	The 3-second playout spread target may not hold across mixed connection quality	Measure across the Phase 1 fleet; if it fails, raise the uniform buffer and accept more latency
T5	Prayer-time computation may disagree with the institution's printed timetable in ways offsets cannot fix	Diff a full month at the first institution before enrolment opens
T6	Carrier CPS limits and CLI-header recognition may make the voice tier answer rate far below 60%	Test with a 30-number list before Phase 2 commits to the tier
T7	Standby power may exceed 0.35 W in practice, making the box conspicuous on a household bill	Measure on revision A; a household that thinks the box costs money will unplug it

T3 is the one most likely to be underestimated. The assumption that a household has Wi-Fi is doing an enormous amount of work in this design, and "has a broadband connection" and "has a reliable 2.4 GHz signal in the room where the box lives" are not the same statement.